



SIRAPID 3344 Steel

Designation by Standards

Brand Name	Ravne	Mat. No.	DIN	EN	AISI/SAE
SIRAPID 3344	BRM3	1.3344	S6-5-3 †	HS6-5-3	M3/2

Chemical Composition (in weight %)

C	Si	Mn	Cr	Mo	Ni	V	W	Others
1.20	max. 0.45	max. 0.40	4.15	4.95	-	2.95	6.35	-

Description

M3 Type 2 is a tungsten molybdenum high speed tool steel designed to provide the great wear resistance required for severe cutting operations. With a raised C and V content. Because of this the steel has optimum durability and cutting performance.

Applications

It is used for high performance mills, reamers, broaches for processing hard materials as well as in the area of cold work steel e.g. for piercing dies in the manufacture of nuts.

Physical properties (average values) at ambient temperature

Density [g/cm³]: 8.07

Coefficient of Linear Thermal Expansion 10⁻⁶ °C⁻¹

20-100°C	20-200°C	20-300°C	20-400°C	20-500°C	20-600°C	20-700°C
11.7	11.9	12.3	12.8	13.0	13.0	13.0

Soft Annealing

Heat to 820-860°C, cool slowly in furnace. This will produce a maximum Brinell hardness of 280.

Hardening

1st pre-heating: 400°C, 2nd pre-heating: 850°C and 3rd pre-heating: 850°C and 1050°C. Harden from a temperature of 1190-130°C followed by oil or air quenching or warm bath at 550°C/air.

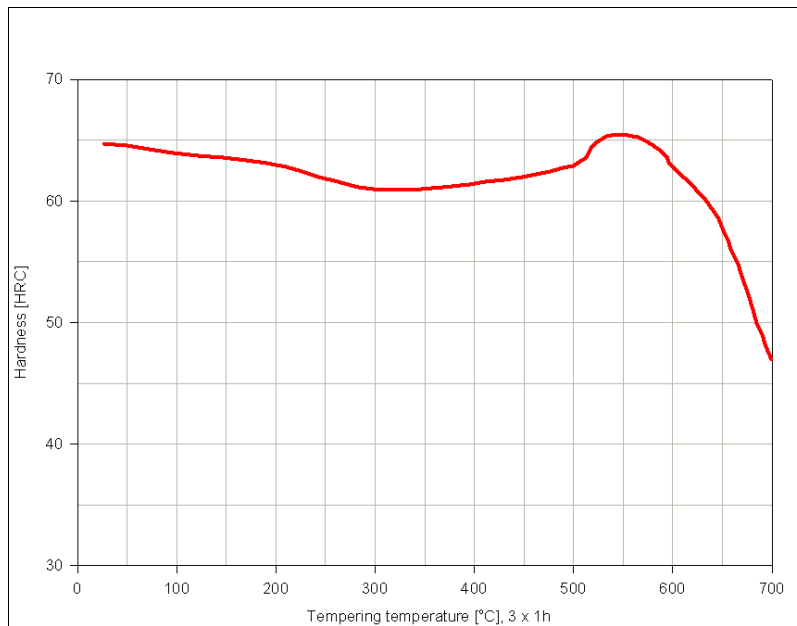
Tempering

Tempering temperature: 540-570°C, at least 3 times. Hardness after tempering is 64-66 HRC.

Tempering Temperature (°C) vs. Hardness (HRC)

200°C	300°C	400°C	500°C	525°C	550°C	575°C	600°C	650°C	700°C
63	61	61.5	63	65	65.5	65	63	58	47

Tempering Diagram



Forging

Hot forming temperature: 1100-900°C.

Machinability

M3 type 2 is classified as a "medium" machinability tool steel in the annealed condition. It may be shaped by grinding but is relatively poor in regard to capability of being ground. Its machinability rating is 50% as compared to the W group water hardening steel.

Corrosion Resistance

Not normally employed in applications requiring corrosion resistance.

Welding

Consult the alloy supplier for information on the advisability of welding.

Forms manufactured: Please see the [Dimensional Sales Program](#).

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